

# Implementation Technology:

## Standard Chips and Programmable Logic Devices

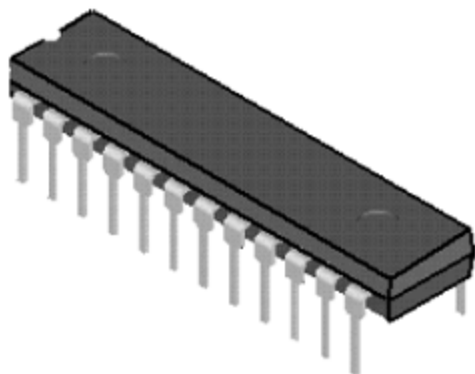
# Standard chips

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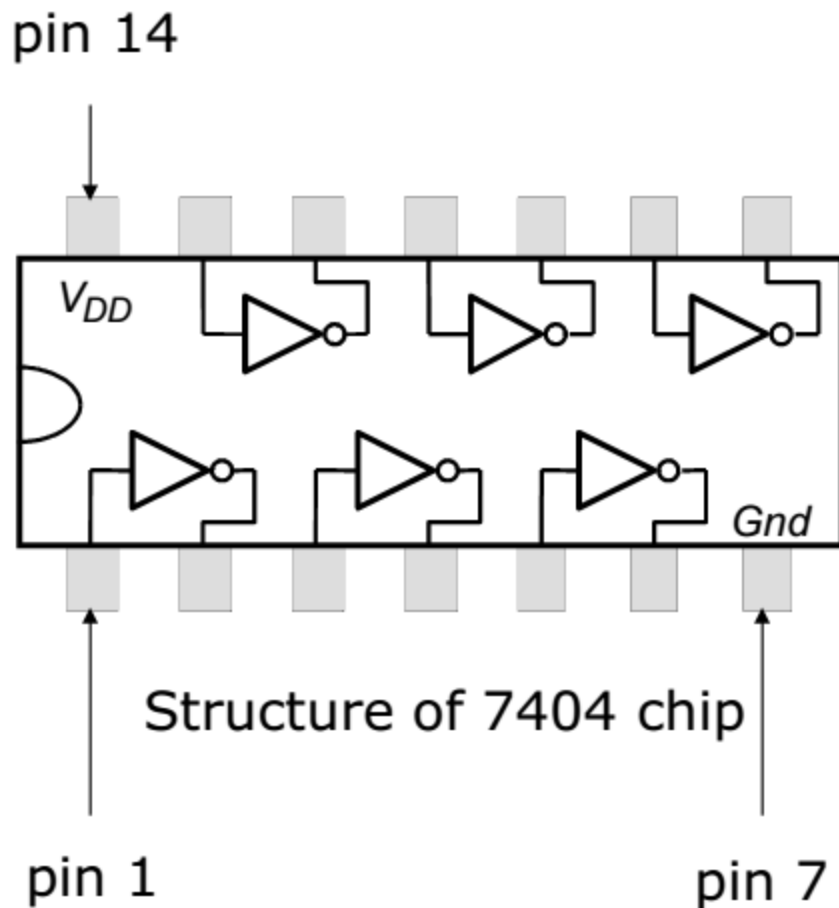
- A number of chips, each with a few logic gates, are commonly used for small logic circuits
- These are known as 7400-series devices because the part numbers always begin with the number 74
  - Commonly packaged in a ***dual-inline package*** (DIP)
  - Chips external connections are called ***pins*** or ***leads***
  - Two pins connect  **$V_{DD}$**  and ***GND*** to supply power for the chip.

# A 7400-series chip

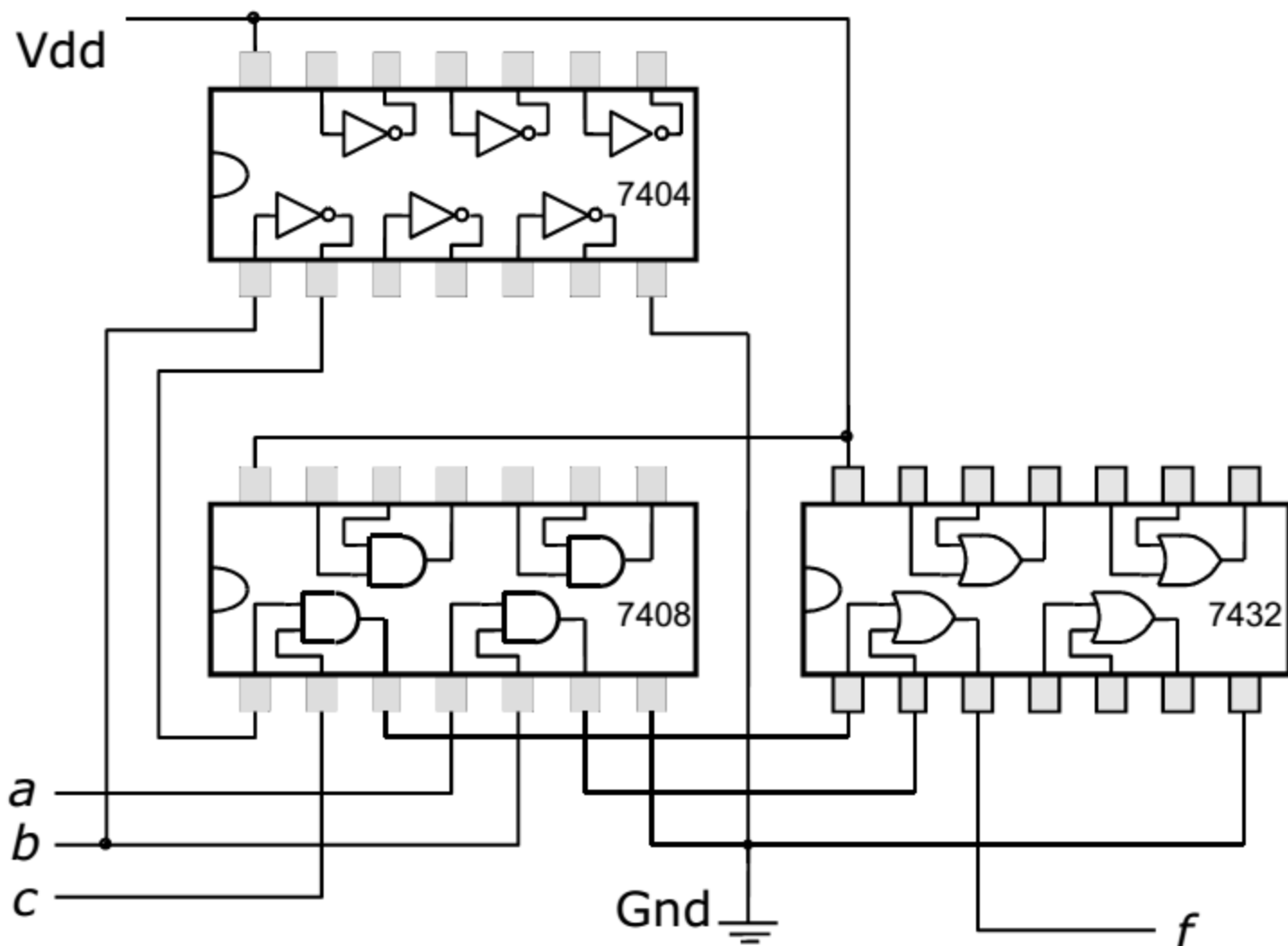
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Dual-inline package



# Implementation of $f=ab+b'c$





## 7400-series chips

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- For each specific 7400-series chip, a number of variants are fabricated with differing technologies
- For example:
  - The 74LS00 is built with a technology called transistor-transistor logic (TTL)
  - The 74HC00 is fabricated using CMOS technology
- Most popular chips in use today are the CMOS variants

# Programmable logic devices

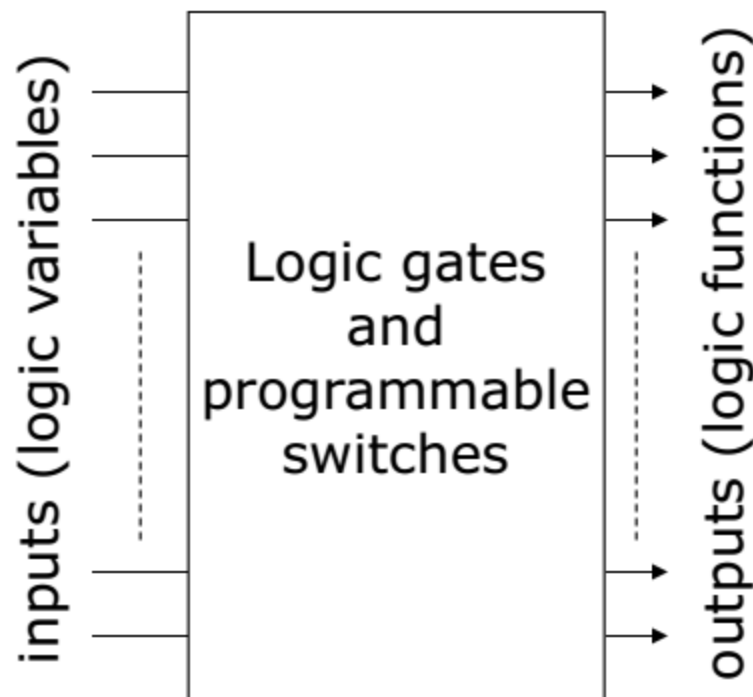
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- The function provided by each 7400-series device is fixed and each chip only provides a few logic gates
  - These limitations make use of these chips inefficient for building large circuits
- It is possible to fabricate chips with a large amount of circuitry (gates) but with a structure (interconnection) that is not fixed
  - Called ***programmable logic devices*** (PLDs)

# Programmable logic devices

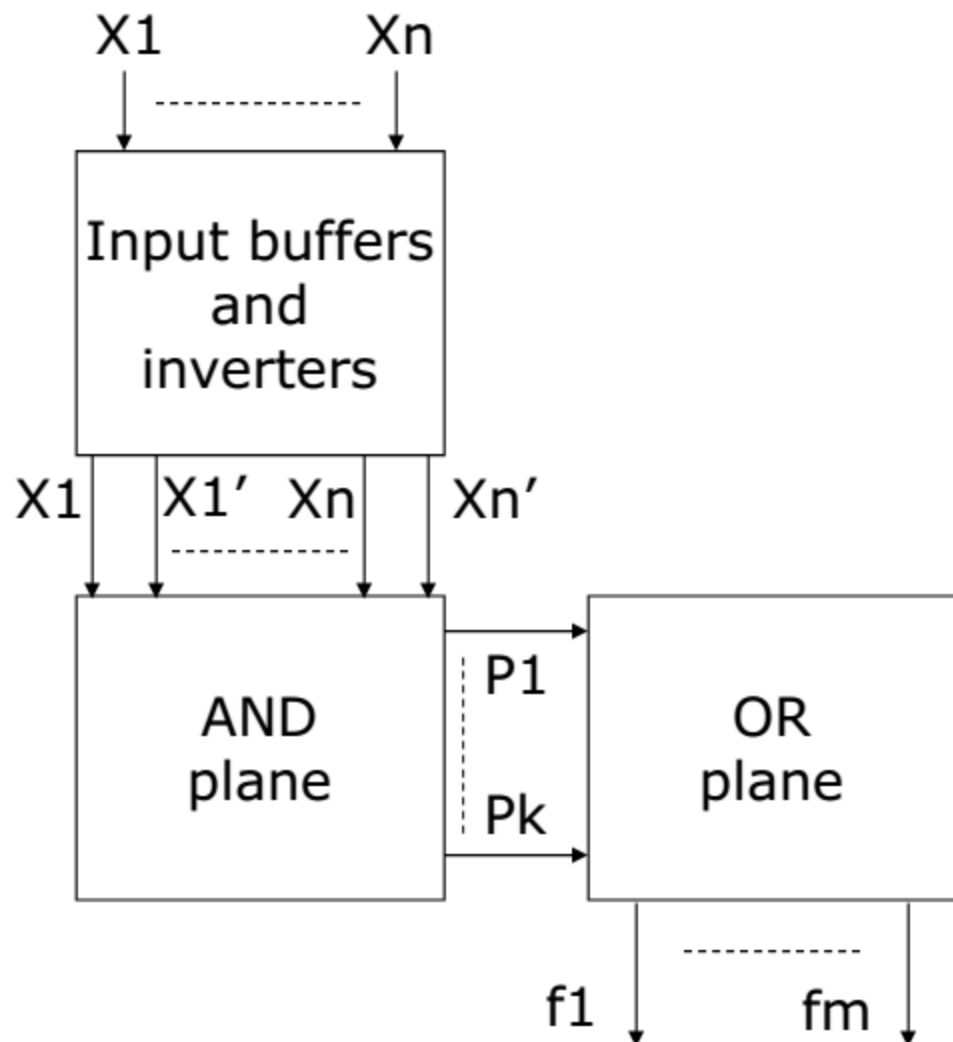
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- A PLD is a general purpose chip for implementing logic circuitry
- Contains a collection of logic circuit elements that can be customized in different ways
- Can be viewed as a black box containing logic gates and programmable switches that allow for different connections between the logic elements
- Can implement whatever logic circuit is needed – subject to limitations of the device



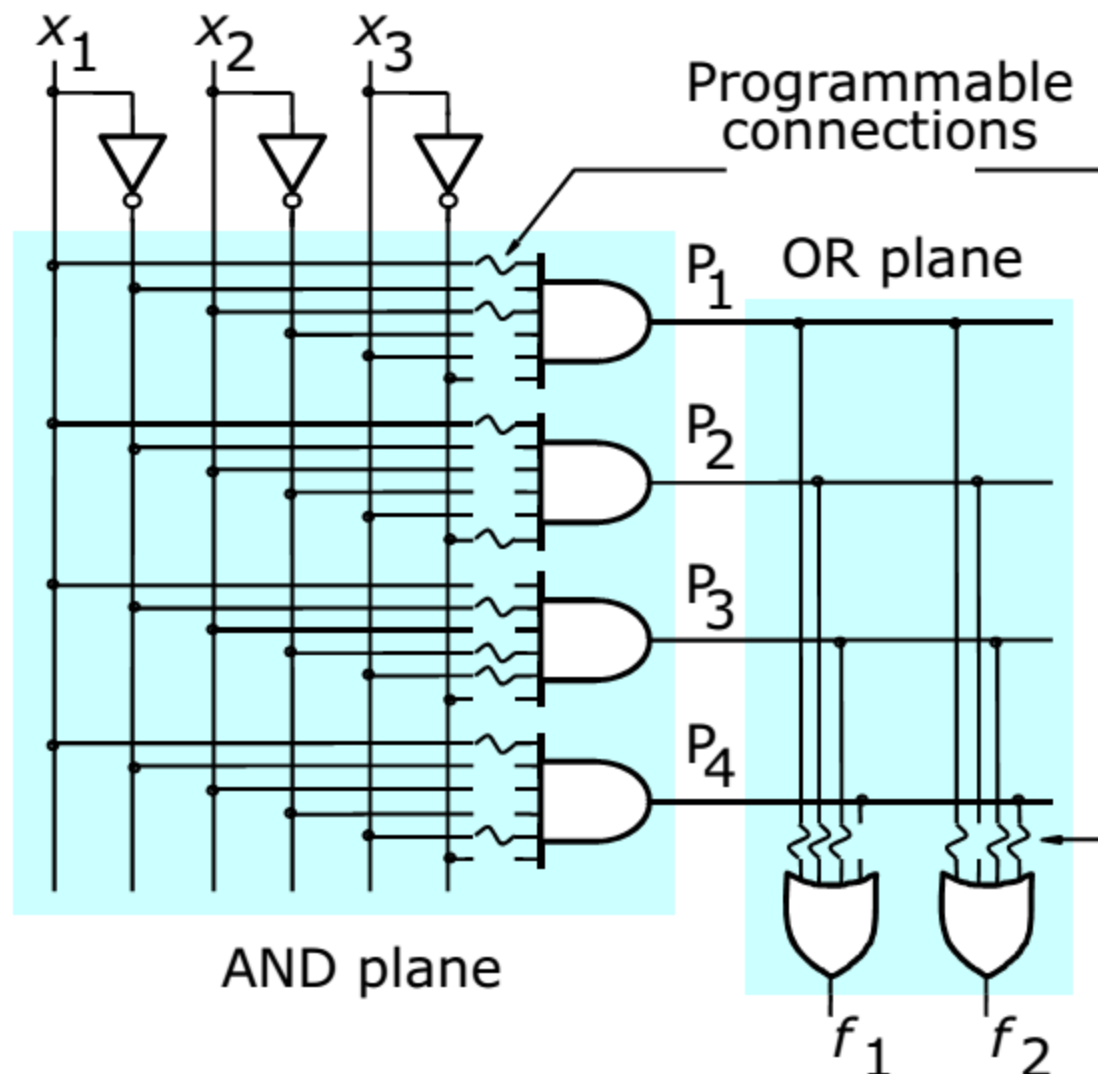
# Programmable Logic Array (PLA)

- The first PLD developed was the **programmable logic array** (PLA)
- Based on the premise that any function can be written in SOP form, a PLA consists of
  - Input buffers and inverters that provide the true and complement form for each input variable
  - A collection of AND gates, with inputs that are selectable (programmable)
  - A collection of OR gates, with inputs that are selectable (programmable)

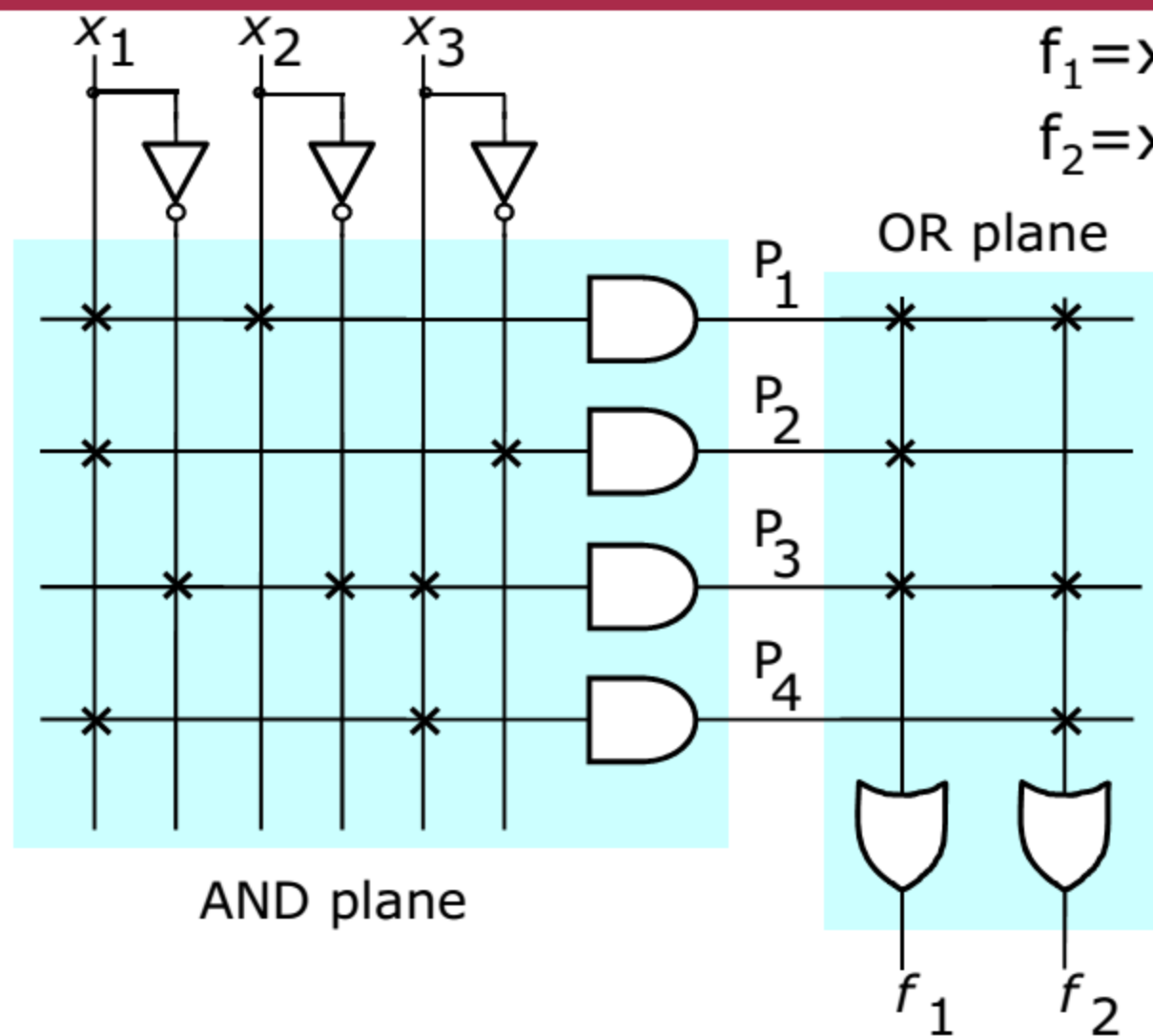




# Gate-level diagram of a PLA



# Customary schematic of a PLA



$$f_1 = x_1x_2 + x_1x_3' + x_1'x_2'x_3$$

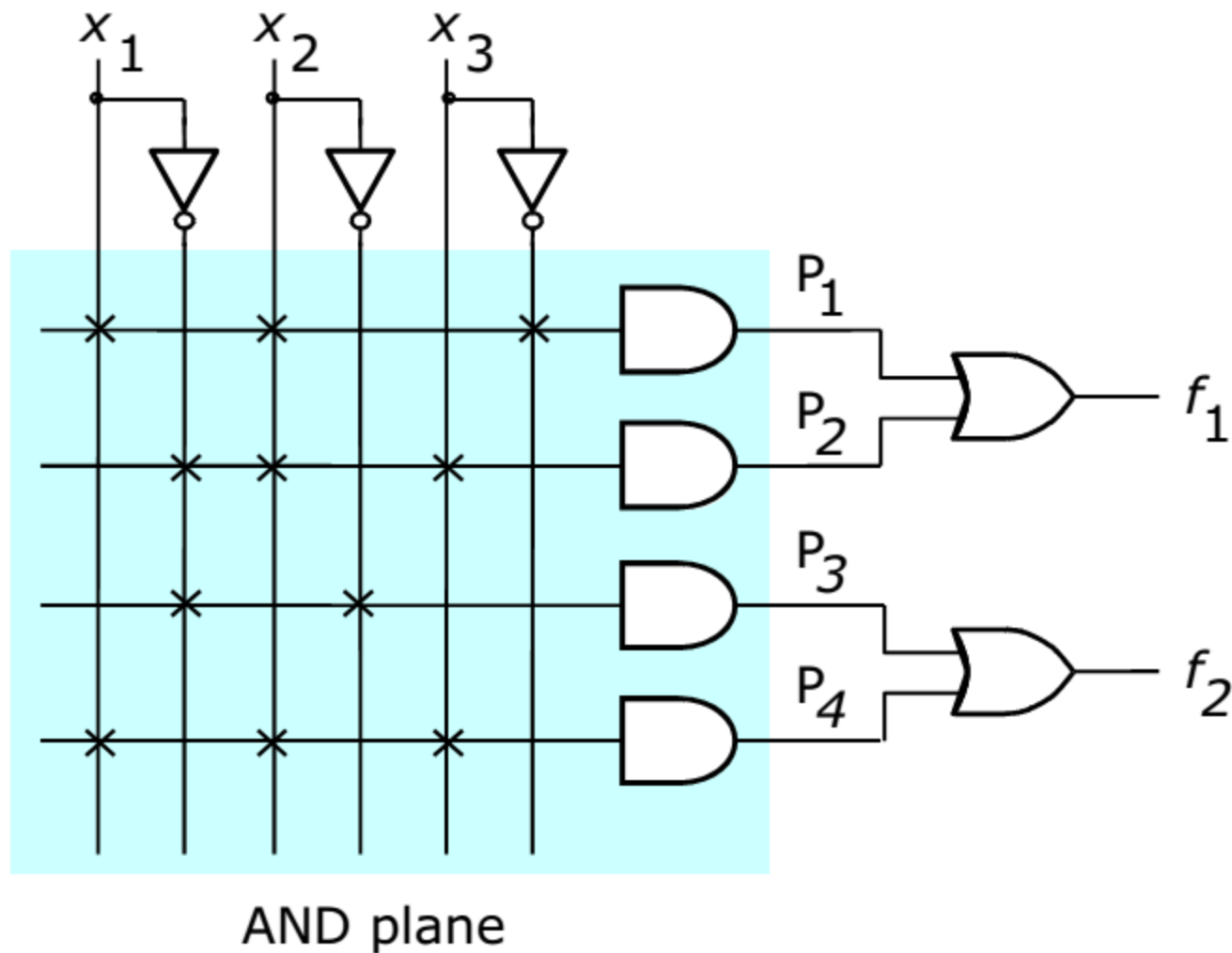
$$f_2 = x_1x_2 + x_1'x_2'x_3 + x_1x_3$$

# Programmable Array Logic (PAL)

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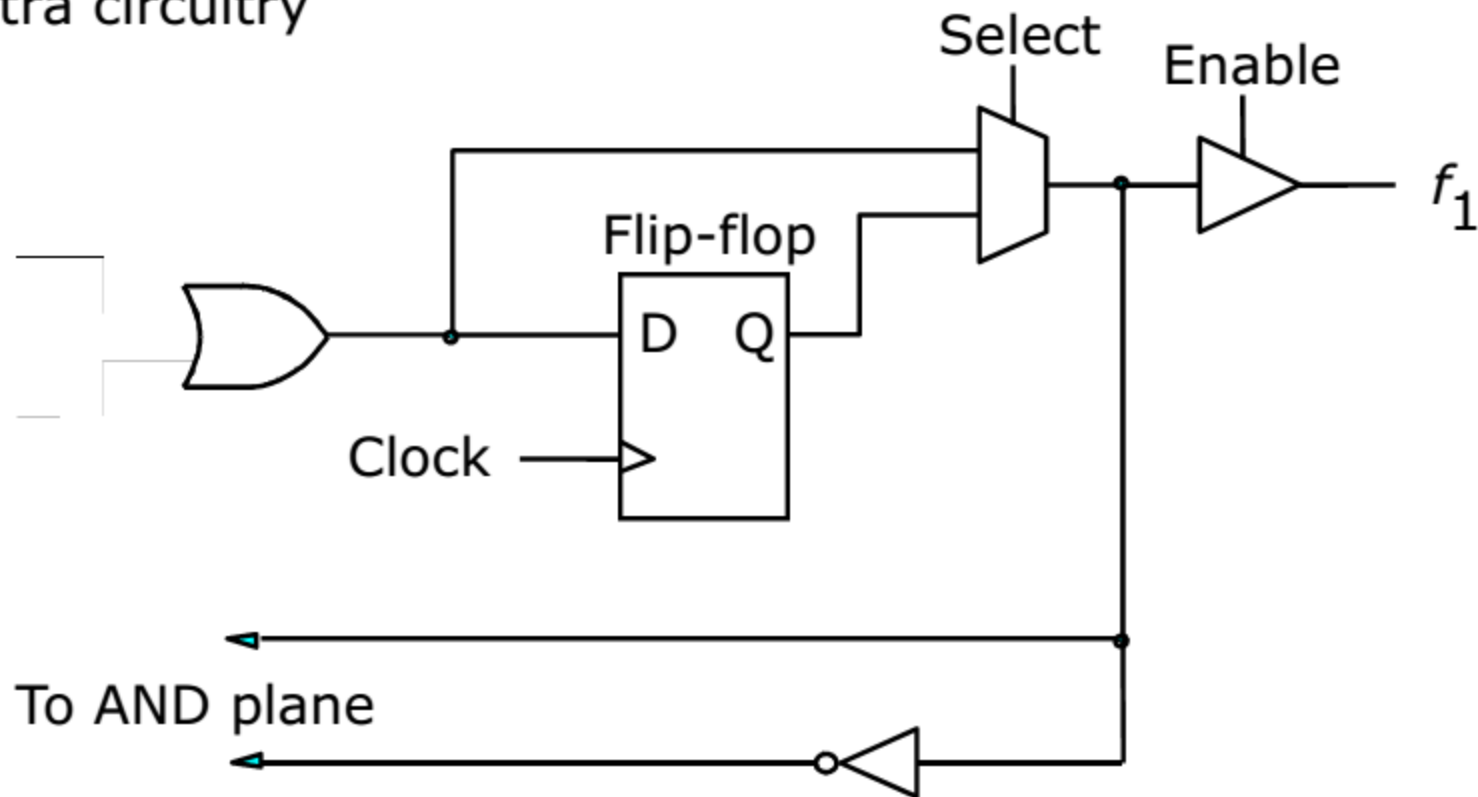
- In a PLA both the AND and the OR planes are programmable
- A simpler device with a fixed OR plane is called a programmable array logic (PAL) device
  - As PALs are easier to manufacture and can operate faster than a PLA, most practical applications using these small programmable devices use the PAL structure

# An example of a PAL



# Extra circuitry in a PAL

- Most actual PAL devices include extra circuitry at the output of each OR gate to provide additional functionality
- The term **macrocell** refers to the OR gate combined with the extra circuitry



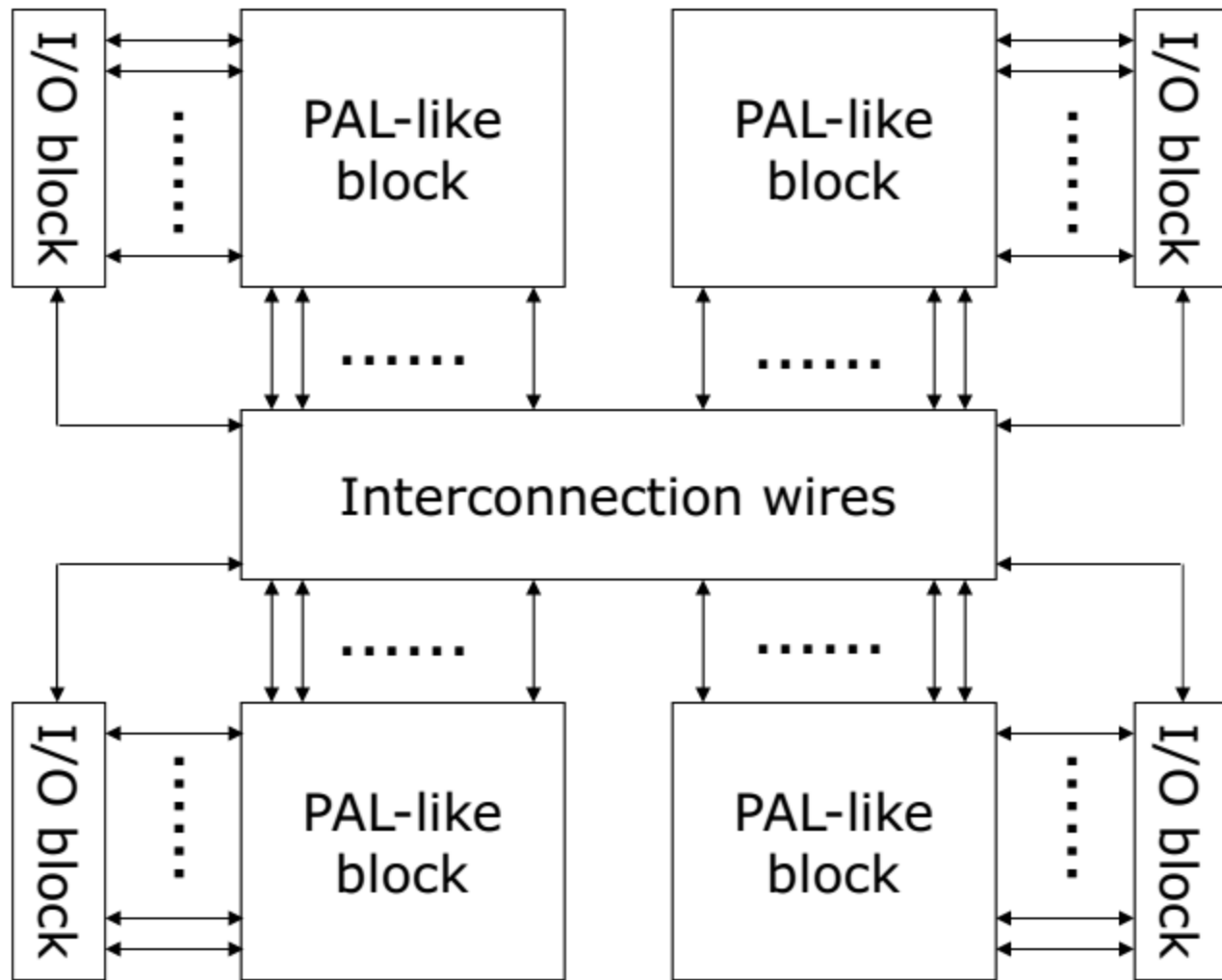
# Complex Programmable Logic Devices (CPLDs)

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- For larger designs that single PLAs or PALs cannot accommodate, a **complex programmable logic device** (CPLD) can be utilized
- A CPLD consists of multiple circuit blocks with internal wiring to connect the blocks together and to the pins on the chip
- Each circuit block is similar to a PAL
  - PAL-like blocks
- Commercial CPLDs have from 2 to more than 100 PAL-like blocks, with 16 macrocells in each block
  - Each macrocell is the equivalent of approximately 20 gates
  - About 20,000 equivalent gates in a CPLD of 1000 macrocells
- Can construct moderately large logic circuits in a single chip

# Structure of a CPLD

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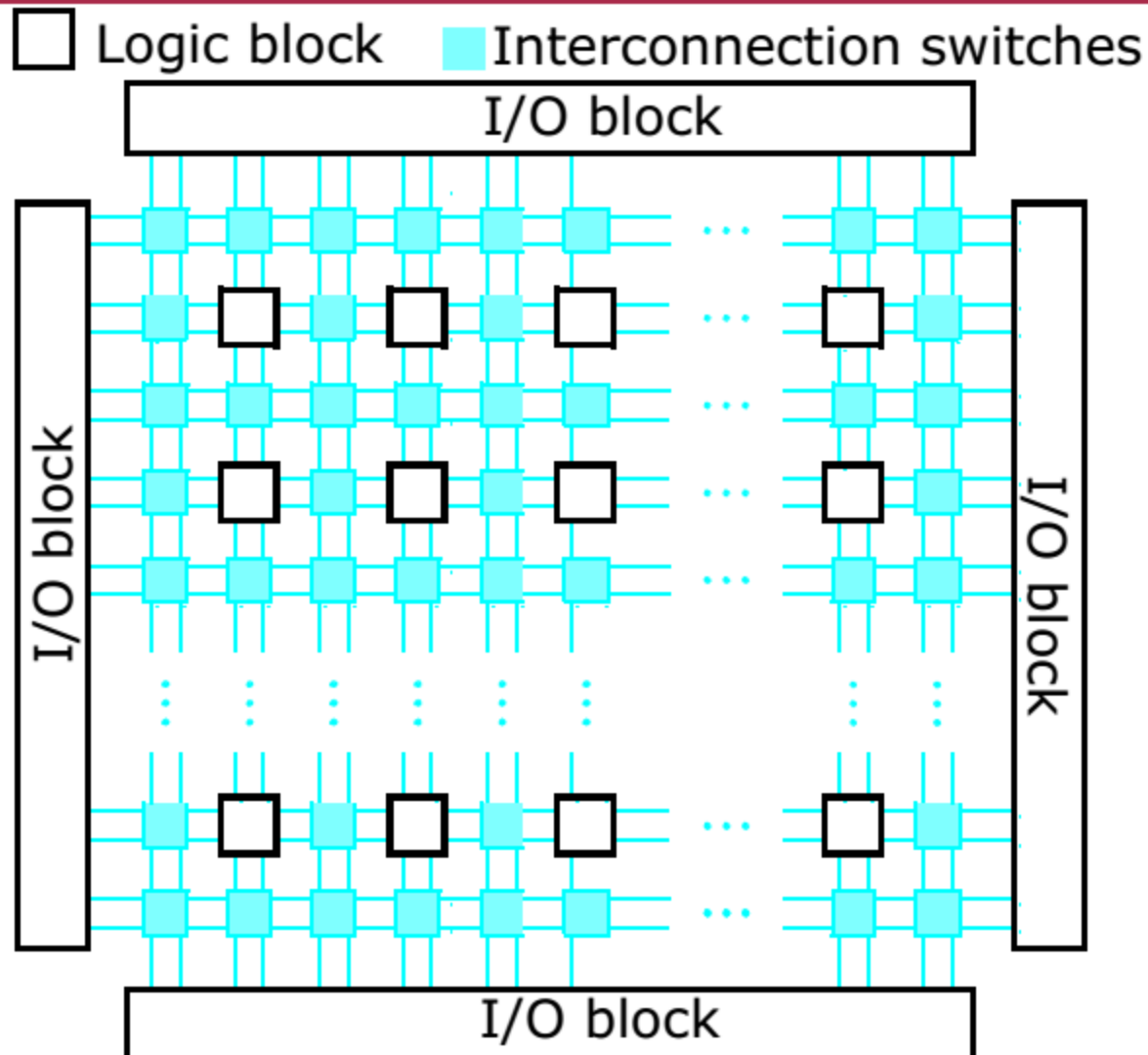
# Field Programmable Gate Arrays

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- To implement even larger circuits, it is convenient to use a different chip that has an even larger logic capacity
  - A field programmable gate array (FPGA)
- Does not contain AND and OR planes
  - Instead provides an array of **logic blocks** and interconnection wires between the logic blocks
  - Interconnection wires are arranged in horizontal and vertical **routing channels** containing wires are programmable switches
- Capable of implementing logic functions of millions of equivalent gates



# Structure of an FPGA



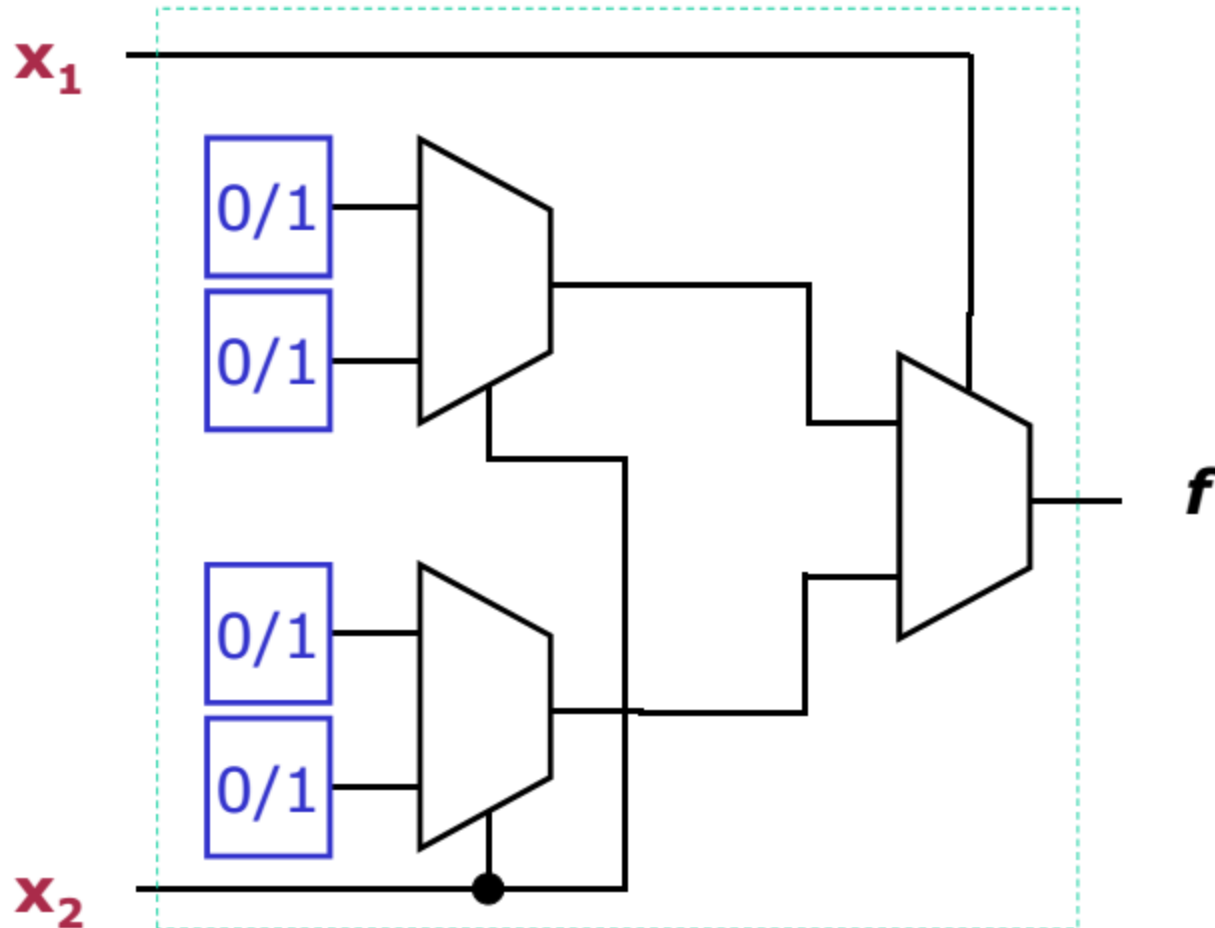
# Look-up tables

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- A logic block commonly used in FPGA devices is the ***look-up table*** (LUT)
- An LUT contains ***storage cells*** that are used to implement small logic functions
- Each cell is capable of storing a single logic value (0 or 1)
- Multiplexers are used to select one of the storage cells for output
- Essentially, the cells store the truth table for a function and the multiplexers select a particular cell for output based on a set of select (control) inputs

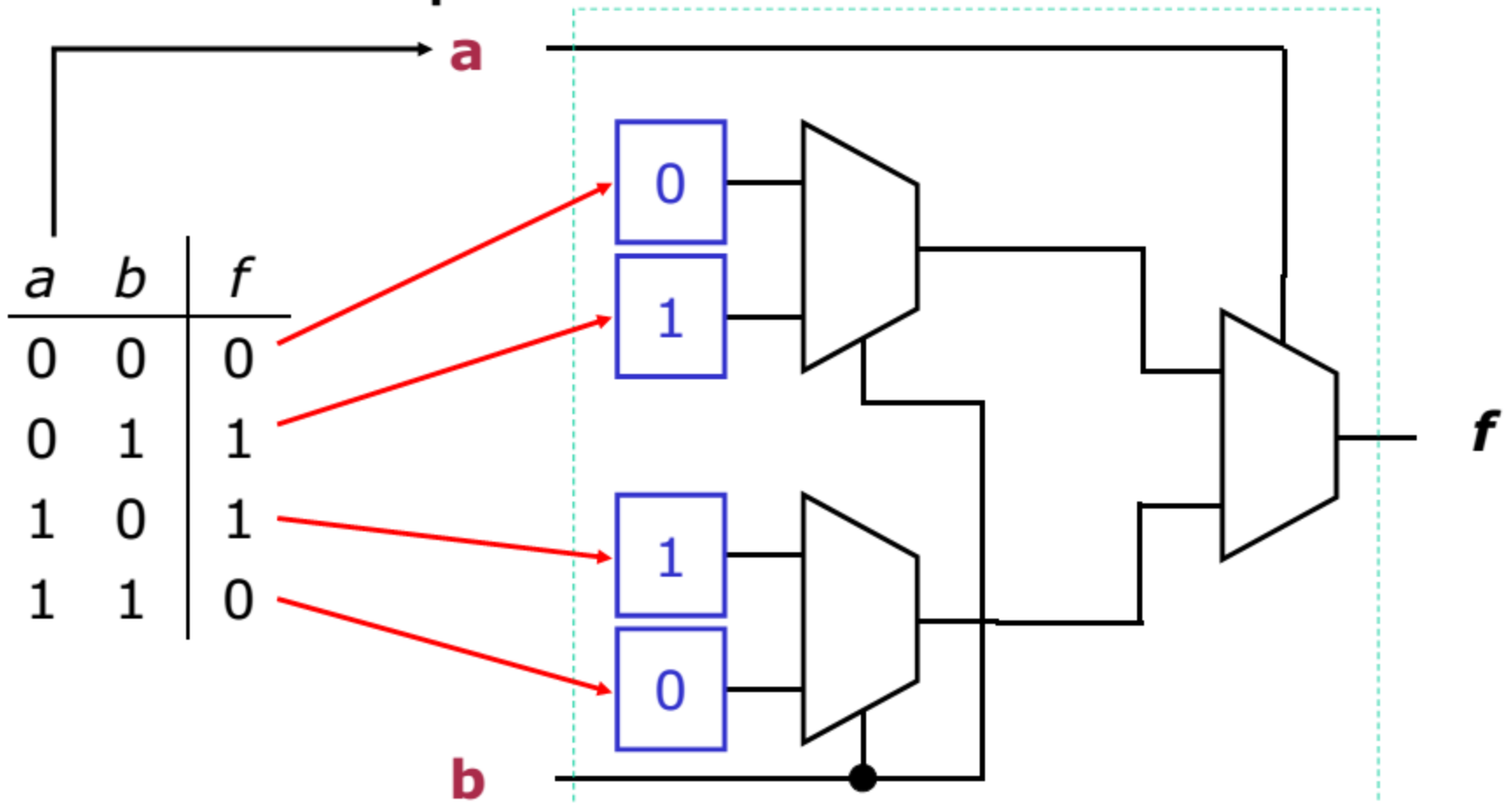
# Two-input LUT structure

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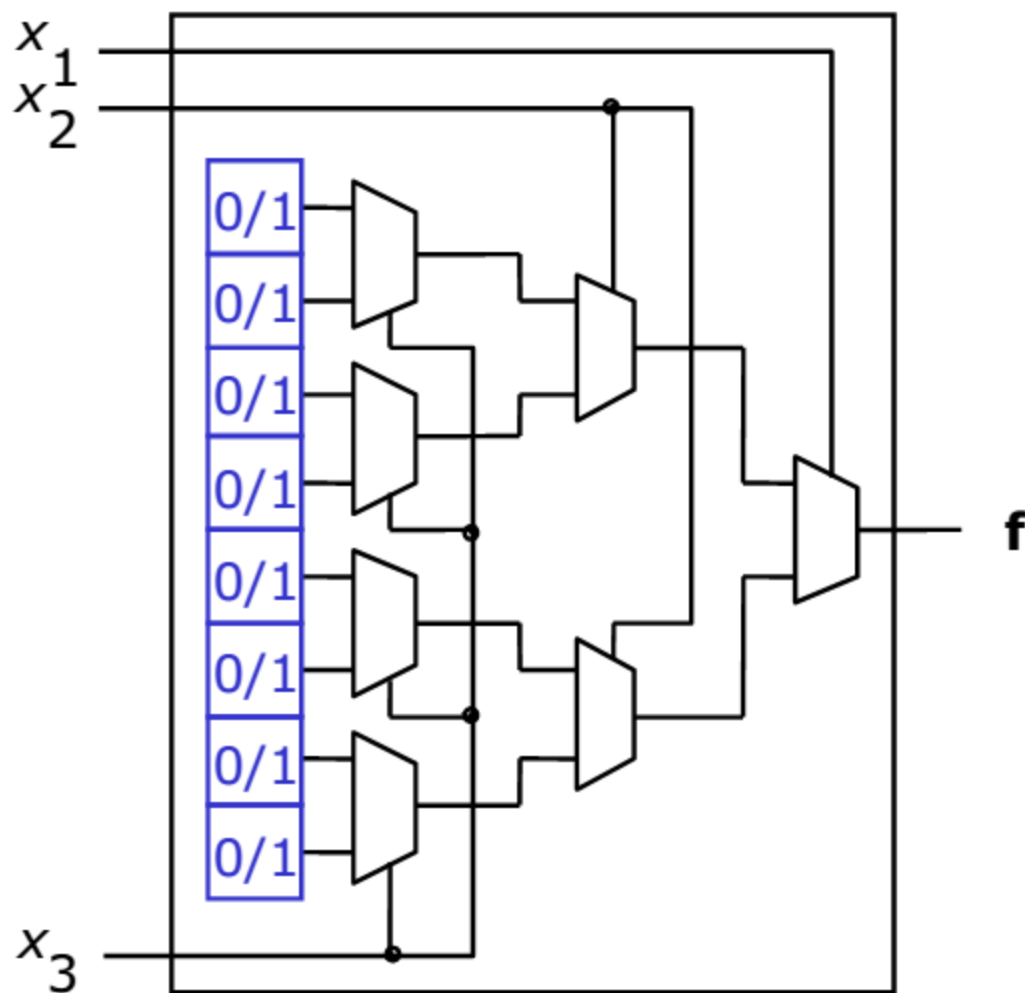
# Programmed LUT ( $f=a'b+ab'$ )

most significant variable  
controls last multiplexer



# Three-input LUT structure

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## Three-input LUT example

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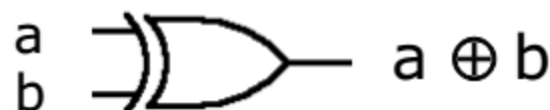
- Show the diagram for a programmed LUT that implements the function
- $f(a,b,c)=a'bc+abc'+ac$

# Exclusive OR (XOR) gate

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- Another basic element, very useful in building circuits that perform arithmetic operations, is the exclusive OR (XOR) gate
- XOR function is denoted with the  $\oplus$  symbol
- In SOP form,  $a \oplus b = ab' + a'b$
- Output is '1' only if the inputs are different

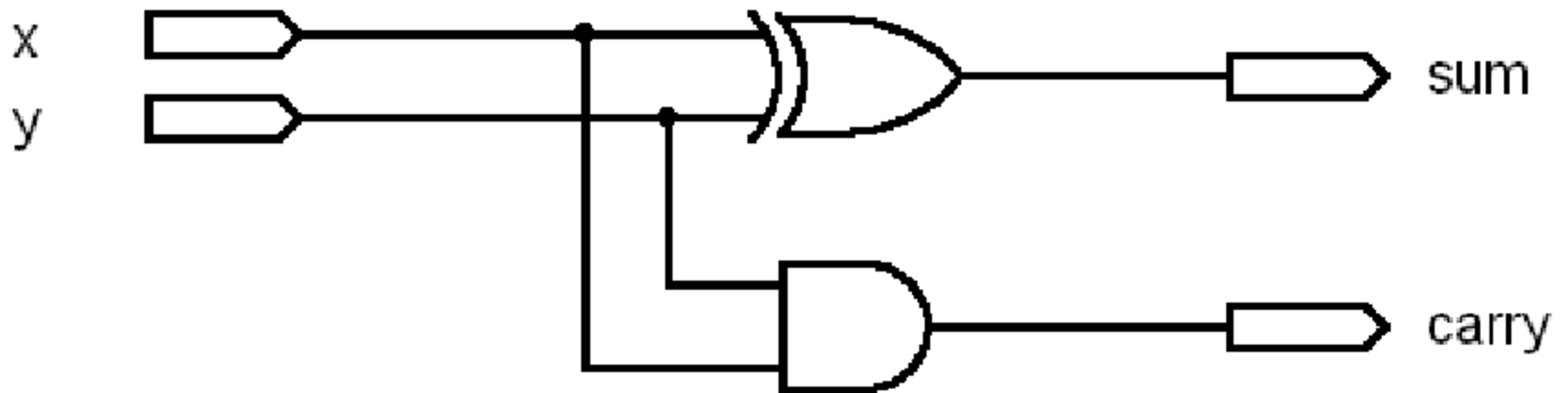
| $a$ | $b$ | $a \oplus b$ |
|-----|-----|--------------|
| 0   | 0   | 0            |
| 0   | 1   | 1            |
| 1   | 0   | 1            |
| 1   | 1   | 0            |



# Example XOR usage

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- Recall the adder circuit
  - $\text{sum} = xy' + x'y$
  - $\text{carry} = xy$





# XOR of three variables

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- What is the canonical SOP form for the following expression?

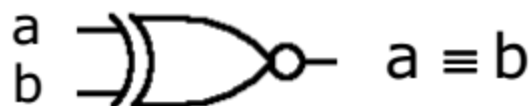
$$f(a,b,c) = a \oplus b \oplus c$$

# Exclusive NOR (XNOR) gate

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- Derived from the XOR function, XNOR is the complement of XOR
- XNOR function is denoted with the  $\equiv$  symbol
- In SOP form,  $a \equiv b = (a \oplus b)' = ab + a'b'$
- Output is '1' only if the inputs are the same
- Also called an equivalence function

| $a$ | $b$ | $a \equiv b$ |
|-----|-----|--------------|
| 0   | 0   | 1            |
| 0   | 1   | 0            |
| 1   | 0   | 0            |
| 1   | 1   | 1            |



# XNOR of three variables

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- What is the canonical SOP form for the following expression?

$$f(a,b,c) = a \equiv b \equiv c$$